Predicting Job Callbacks Using Resume Data

A Statistically-Sound Data Science Approach Using Machine Learning

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Abstract

This project explores how data science, specifically statistical testing and machine learning models, can be used to analyze factors influencing job acceptance and predict whether a job applicant receives a callback based on their resume. We begin with a structured resume dataset, clean and validate the data, and engineer features that capture age, experience, education level, and other career attributes. After ensuring data quality through rule-based filters, the data is used to train a machine learning model, specifically both a Logistic Regression model and Random Forest classifier to predict callback outcomes.

The model is evaluated using Accuracy and the Area Under the Receiver Operating Characteristic curve (ROC-AUC). The Random Forest model achieves a comparatively superior accuracy of 70.54% and a ROC-AUC of about 0.81, suggesting successful discrimination. Dimensionality reduction using Principle Component Analysis (PCA) is also performed to better understand structure in the high dimensional feature space created by one hot encoding. Visual analysis in Colab, such as age and experience distributions, callback rates by education level and experience group, and a ROC curve, help interpret the patterns behind callback decisions. Additionally, K-Prototypes clustering is performed to construct detailed profiles of successful candidates, providing analysis of the “ideal” applicant profiles.

Because the dataset is artificial and many attributes are randomly generated, the results are not evidence about real world hiring practices. Instead the goal is to demonstrate a statistically sound data science workflow for resume analysis: cleaning and validating data, exploring relationships through visualization and hypothesis testing, building machine learning models, and interpreting results through clustering and dimensionality reduction.

Introduction

The modern job market is increasingly competitive; in recent years, job postings have fallen by 13% in Texas (Jennings 2025). Both applicants and employers rely on digital tools to manage large volumes of resumes. For example, over 65% of recruiters have already implemented AI-powered techniques. 44% of recruiters have moved to these techniques to save time, 58% to improve candidate sourcing (both candidates who fit the position and are likely to accept), and to reduce hiring costs by up to 30% per hire (Kumar 2023). At the same time, there is growing concern about fairness, transparency, and the potential for algorithmic bias in automated hiring. Understanding how resume characteristics translate into callback is therefore important for both data scientists and job seekers. Data science techniques can streamline recruitment using prediction-based insights and candidate clustering. Data science allows for objective analysis, but can sometimes exist in a black box for candidates and recruiters alike. Therefore, this project helps provide insight for all levels of the recruitment process.

Description of Dataset

The dataset used in this project, entitled “Job & Resume Matching Dataset”, was sourced from Kaggle. This dataset provides common attributes from applicants included on their resume in a highly structured format. It is important to note that this dataset is likely artificial, as detailed by the random distribution of all features, and the unrealistic structure of the data (for example, having only one “skill” included per resume). The initial file has been updated in the following ways:

1. Saved as a CSV file for easy manipulation and compatibility for further analysis.
2. For purposes of this assignment, artificial job acceptance labels have been randomly assigned to the resume entries.
3. Removed nonsensical columns (cover letter, LinkedIn profile link, GitHub profile link, volunteer experience, and certifications.

Further information on the attributes in this dataset is provided in **Table 1**, which is generated from the dataset itself, the Kaggle landing page, and initial Python analysis**.** It is important to note that some of the exact measurements and meanings of the attributes in this dataset were unclear and had to be extrapolated.

Table 1: Column Titles, Data Types, and Descriptions of Dataset Attributes

Attribute Title	Data Type	Description
first_name	object	First name of the applicant
last_name	object	Last name of the applicant
age	int64	Age of the applicant in years
email	object	Email address of the applicant
country	object	Country where applicant

		currently resides
education_level	object	Description of education level of the applicant
skills	object	Singular skill, often category, of applicant
experience_years	int64	Number of years of experience of applicant
current_job_title	object	Title of applicant's current position
languages_spoken	object	Singular language, spoken by applicant
technical_proficiencies	object	Singular technical proficiency, often a coding language or technique, of the applicant
references_available	bool	True/False if references available for applicant
availability_date	object	Date the applicant is available to begin new position
github_contributions	int64	Number of GitHub contributions linked to applicant
previous title level	object	Description of level/title of previous job

professional level	object	Category of applicant's professional level
additional_training	object	Singular additional training received, often category, of the applicant
job_accepted	int64	1 = accepted to job applied to 0 = rejected from job applied

Data Cleaning

Various logical checks were performed in order to clean the data and make the artificial data as representative of real trends and resumes as possible. The following rules were applied:

1. If an applicant shows a greater number of years of experience than their age, they are removed from the dataset. This is logically impossible.
2. If the applicant started working before age 16 (difference between age and years of experience is less than 16), they are removed from the dataset. This is to represent child labor laws.
3. If the applicant has a negative age or negative years of experience, they are removed from the dataset. This is to represent real-world scenarios.
4. If an applicant has more than 400 GitHub contributions, they must have at least 2 years of experience or a Master's/PhD; otherwise, they are removed. (Harding)
5. If an applicant has more than 700 GitHub contributions, they must have at least 4 years of experience or a Master's/PHD; otherwise, they are removed. (Harding)
6. An applicant must meet a specific age requirement by professional level in order to ensure that these are plausible combinations in applicants.
    a. Junior level = minimum age 18
    b. Mid level = minimum age 25
    c. Senior level = minimum age 30
    d. Lead level = minimum age 32
    e. Executive = minimum age 40

7. An applicant must meet a specific number of years of experience by professional level in order to ensure that these are plausible combinations in applicants.
    a. Junior level = minimum 0 years of experience
    b. Mid level = minimum 2 years of experience
    c. Senior level = minimum 5 years of experience
    d. Lead level = minimum 8 years of experience
    e. Executive = minimum 12 years of experience
8. An applicant must meet a specific age requirement by educational level in order to ensure that these are plausible combinations in applicants.
    a. Associate's degree = minimum age 19
    b. Bachelor's degree = minimum age 21
    c. Master's degree = minimum age 23
    d. PhD = minimum age 26
    e. Technical degree = minimum age 18
    f. Professional certification = minimum age 18
    g. No degree = minimum age 18 (as a standard check; no applicants below age 18)
    h. Trade school certificate = minimum age 19
    i. Some college = minimum age 18
    j. High school diploma = minimum age 17 (as a standard check; no applicants below age 18)
    k. GED = minimum age 17 (as a standard check; no applicants below age 18)

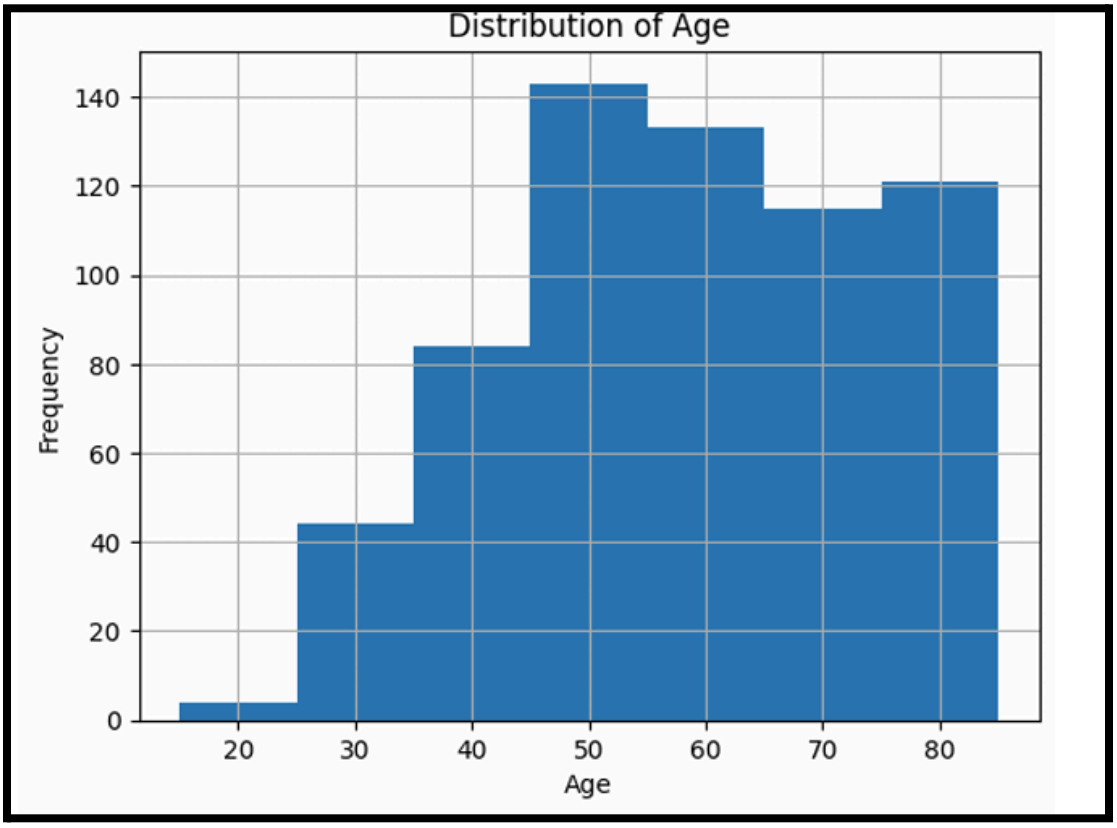
Additionally, whitespace was removed and all text-based attributes were converted to lowercase in order to ensure consistency and allow for meaningful future analysis. All entries with empty characteristics and all entries that were exact duplicates were also eliminated. After applying these rules, the initial 1000 entries were pared down to 644 valid entries for further analysis.

Future rules could include removing unlikely combinations, such as 15+ years of experience with a higher education degree and in a lower-level position, and removing people past the standard retirement age.

Exploratory Data Analysis & Knowledge Discovery

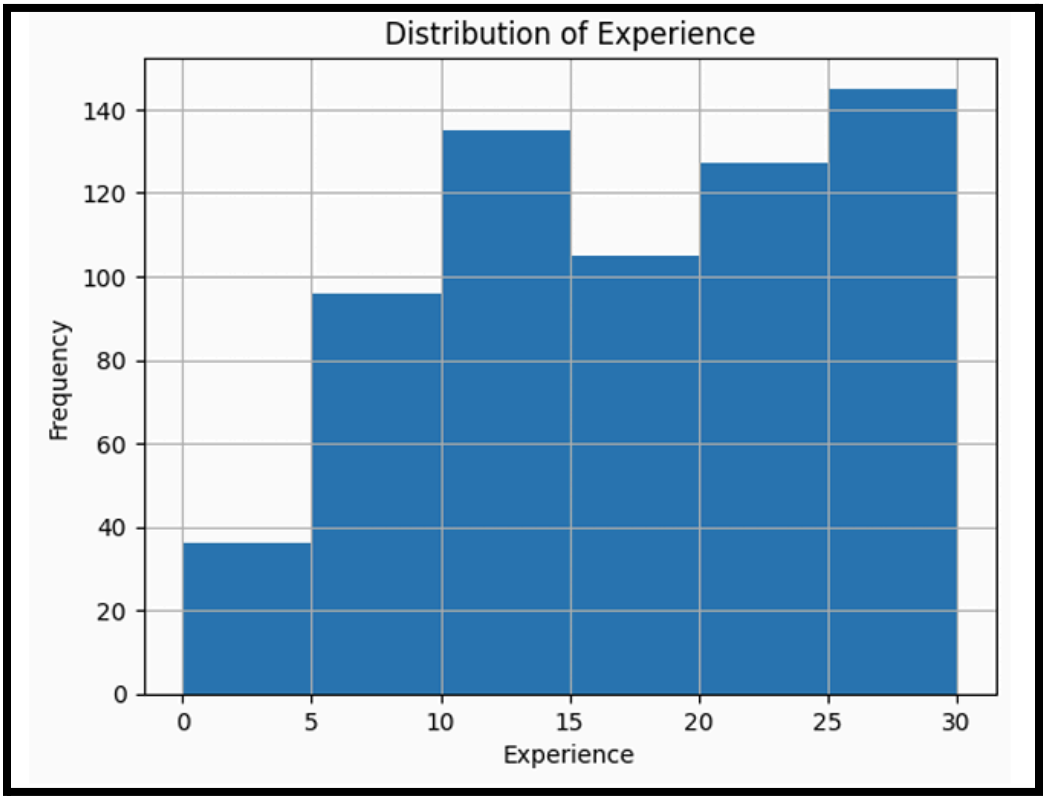
Various visualizations and statistical tests were performed in order to gain further knowledge and insight into the construction of this data and patterns that may emerge.

Graph 1: Histogram of Age Distribution



The above image displays the frequency (count of candidates) across different age ranges (in years). The data has been divided into 7 bins of 10 year age ranges. The least frequent age range is the youngest candidates, from age 15 to age 25. The two most represented age groups are from 44-55 and 55-65, showing older, likely more experienced candidates. This logical trend matches the distribution of experience shown in **Graph 2**, but can introduce bias into the further analysis. The data is not normally distributed, showing an almost uniform distribution across the older candidates.

Graph 2: Histogram of Experience Distribution

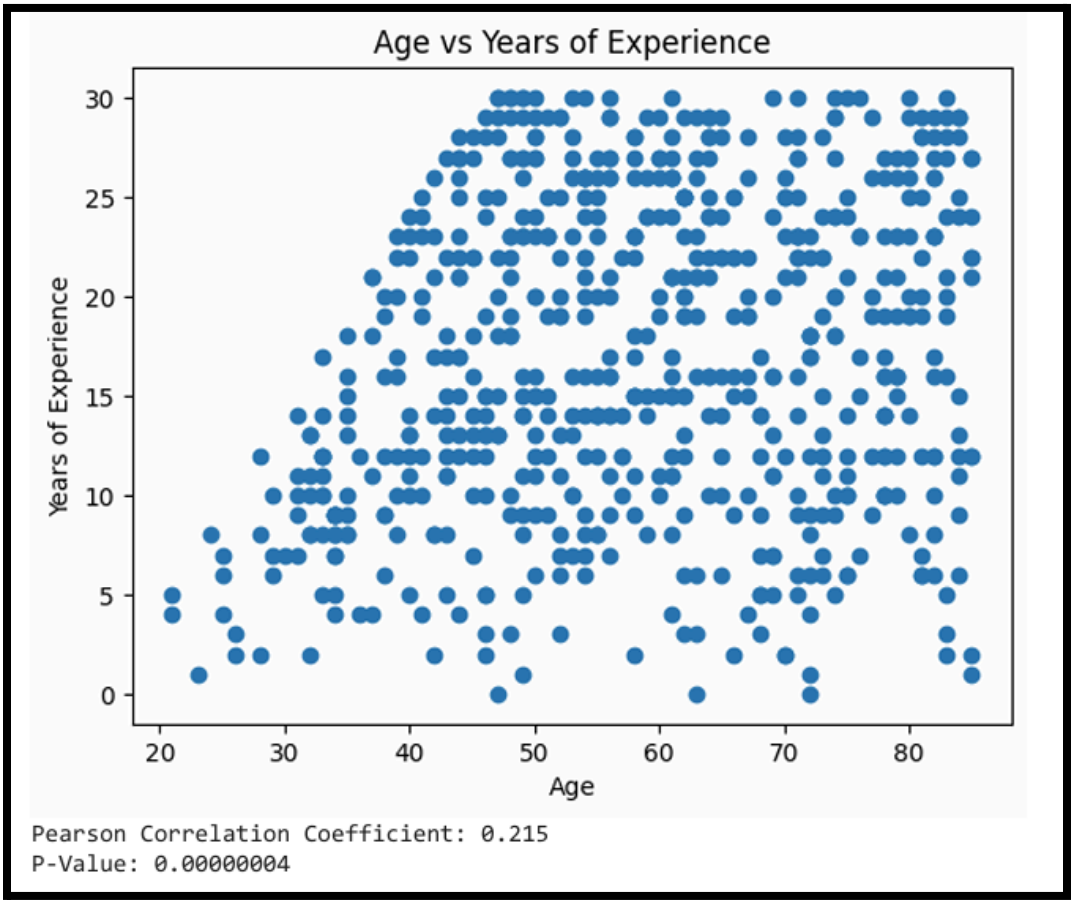


The above image displays the frequency (count of candidates) across different ranges of professional experience (in years). The data has been divided into 6 bins of 5 year intervals. The least experienced candidates (0-5 years) are the least common in the dataset. The most common experience range is the most experience (25-30 years). The second most common range of experiences is in the mid-range of 10-15 years of experience. This is not normally distributed data, as it shows a multi-model tendency. The dataset is more heavily weighted towards candidates with longer careers, which can introduce bias into the result. This matches with the pattern shown in **Graph 1** of having more mature candidates represented.

Correlation Tests

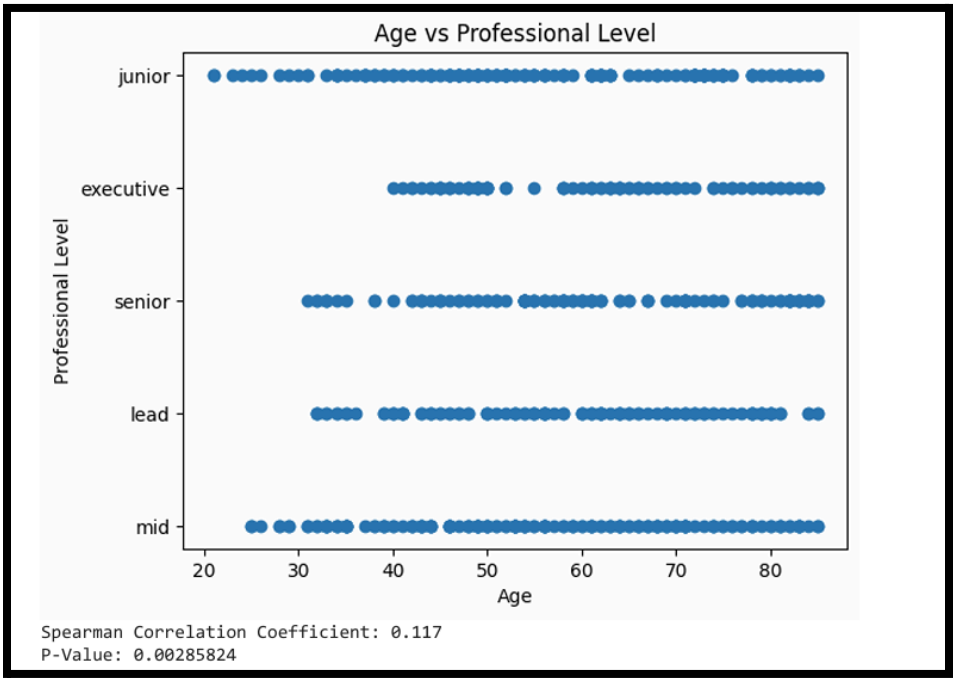
Various variables, that logically might display some form of correlation (especially after being filtered previously), were tested for correlation in this initial data analysis using the Pearson Correlation Coefficient and the Spearman Correlation Coefficient ("Pearson Correlation Test Between Two Variables - Python") where relevant.

Graph 3: Scatter Plot of Age vs Years of Experience



The above graph shows age in years being plotted against experience in years. There are very few data points in the upper left-hand corner, which represents low age, high experience combinations, which is logical and expected. This was accounted for with the resume filtering. However, besides that section, the data points are very scattered without much in terms of clustering or trends. The Pearson Correlation test shows a weak positive correlation that as age increases, years of experience increase; this is as expected, but might anticipate a stronger correlation. The P-value is less than 0.05, showing that this is a statistically significant relationship. It is of note that if pre-processing had not been performed, there would likely have either been a weaker relationship or statistically insignificant relationship.

Graph 4: Scatter Plot of Age vs Professional Level



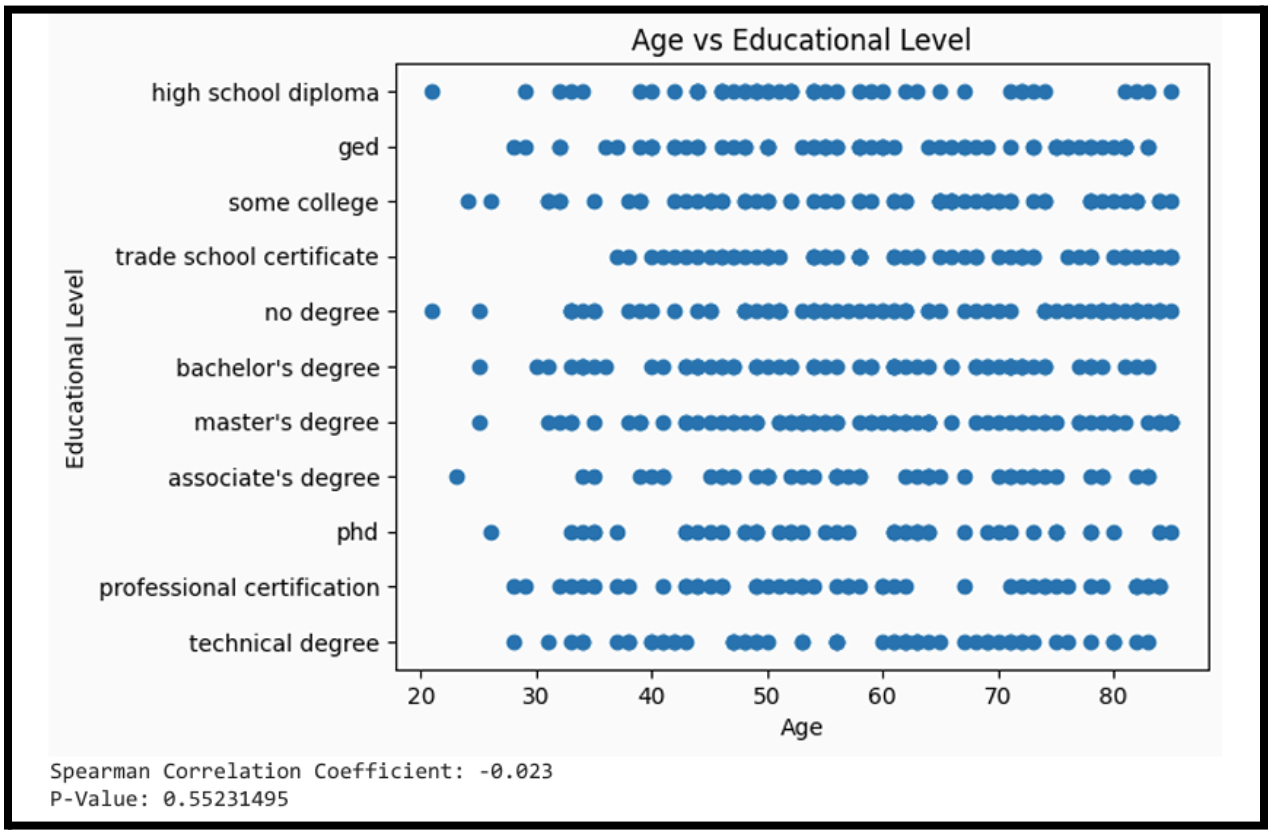
The above graph shows age in years being plotted against professional level by category. There are very few data points for younger candidates outside of the junior category, which is to be expected with the previous logic checks. However, besides that section, the data points are very uniformly distributed without much in terms of clustering or trends. The Spearman Correlation test was performed, as this can be used to check continuous variables like age against an ordinal variable like professional level. For the sake of this analysis, the professional levels were converted to numerical categories as follows:

Table 2: Numerical Mapping of Professional Levels

Numerical Mapping	**Professional Level**
1	Junior
2	Mid
3	Senior
4	Lead
5	Executive

The Spearman Correlation test shows a weak positive correlation that as age increases, so does professional level; this is as expected, but might anticipate a stronger correlation, as this relationship is nearly negligible. The P-value is less than 0.05, showing that this is a statistically significant relationship. It is of note that if pre-processing had not been performed, there would likely have either been a weaker relationship or statistically insignificant relationship.

Graph 5: Scatter Plot of Age vs Educational Level



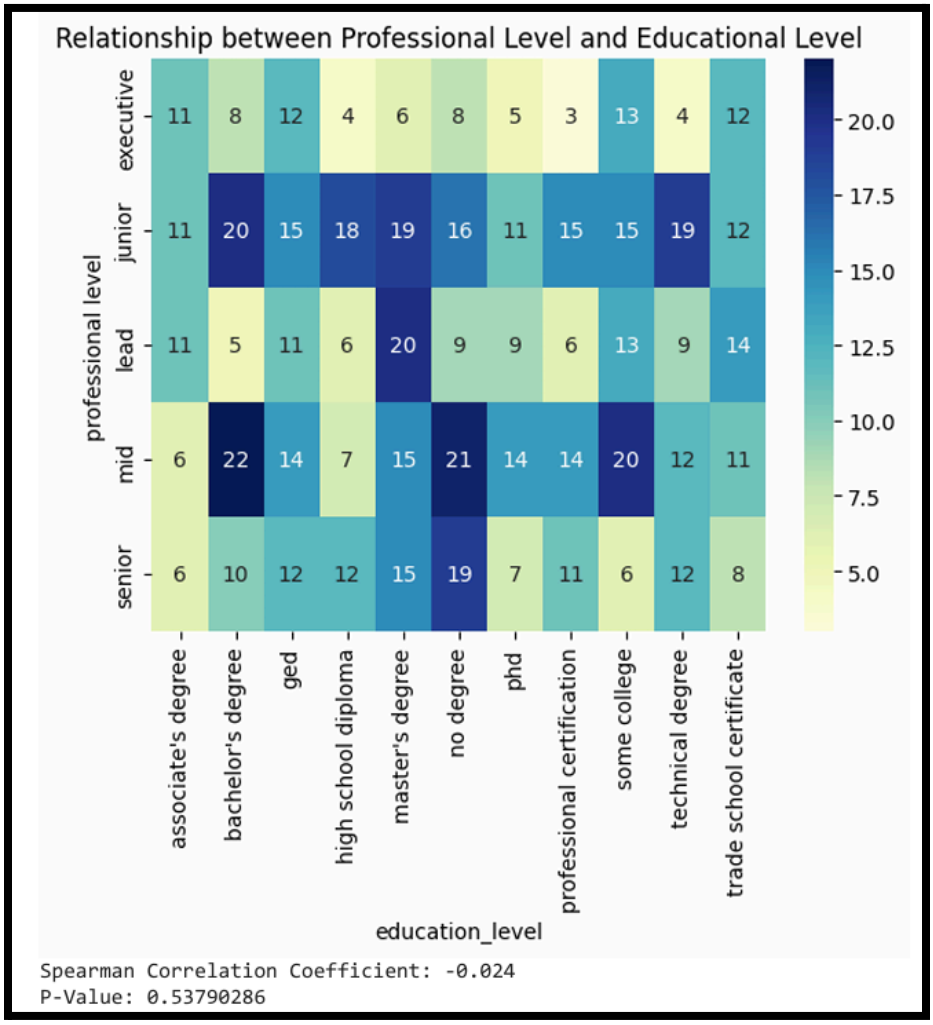
The above graph shows age in years being plotted against educational level by category. The data points are very uniformly distributed without much in terms of clustering or trends. The Spearman Correlation test was performed, as this can be used to check continuous variables like age against an ordinal variable like educational level. For the sake of this analysis, the educational levels were converted to numerical categories as follows:

Table 3: Numerical Mapping of Educational Levels

Numerical Mapping	Educational Level
0	No Degree
1	GED
2	High School Diploma
3	Trade School Certificate
4	Some College
5	Professional Certification
6	Technical Degree
7	Associate's Degree
8	Bachelor's Degree
9	Master's Degree
10	PhD

The Spearman Correlation test shows a weak negative correlation that as age increases, educational level decreases. This can match current trends in increasing higher education, though sometimes with its value not inflating as well. However, this relationship is so close to zero as to be completely negligible. In addition, the P-value is greater than 0.05, meaning that this relationship is not statistically significant and could also occur in this manner due to chance. This data was likely generated randomly, so this result is to be expected, despite the pre-processing measures taken.

Graph 6: Heatmap of Educational Level Against Professional Level



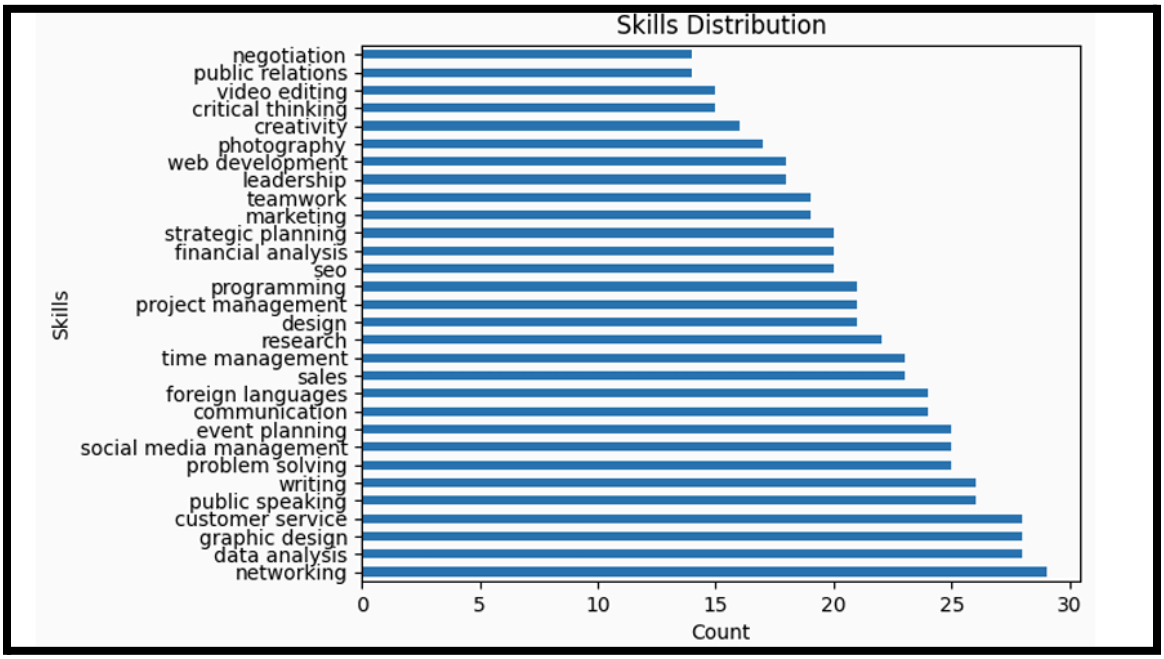
The above graph shows educational level by category plotted against professional level by category in a heatmap, which is useful for showing the strength of the "overlap" and correlation for two categorical/ordinal variables ("How to create a correlation heatmap in Python?"). The data points are very uniformly distributed without much in terms of clustering or trends. The Spearman Correlation test was performed, as this can be used to check continuous variables against each other. The numerical mapping as described for use in **Table 2** and **Table 3** were employed here as well. The Spearman Correlation test shows a weak negative correlation that as educational level increases, professional level decreases. This can match current trends in increasing higher education, though sometimes with its value not inflating as well. However, this relationship is so close to zero as to be completely negligible. In addition, the P-value is greater than 0.05, meaning that this relationship is not statistically significant and could also occur in

this manner due to chance. This data was likely generated randomly, so this result is to be expected, despite the pre-processing measures taken.

Skills, Technical Proficiencies, and Training by Professional Level

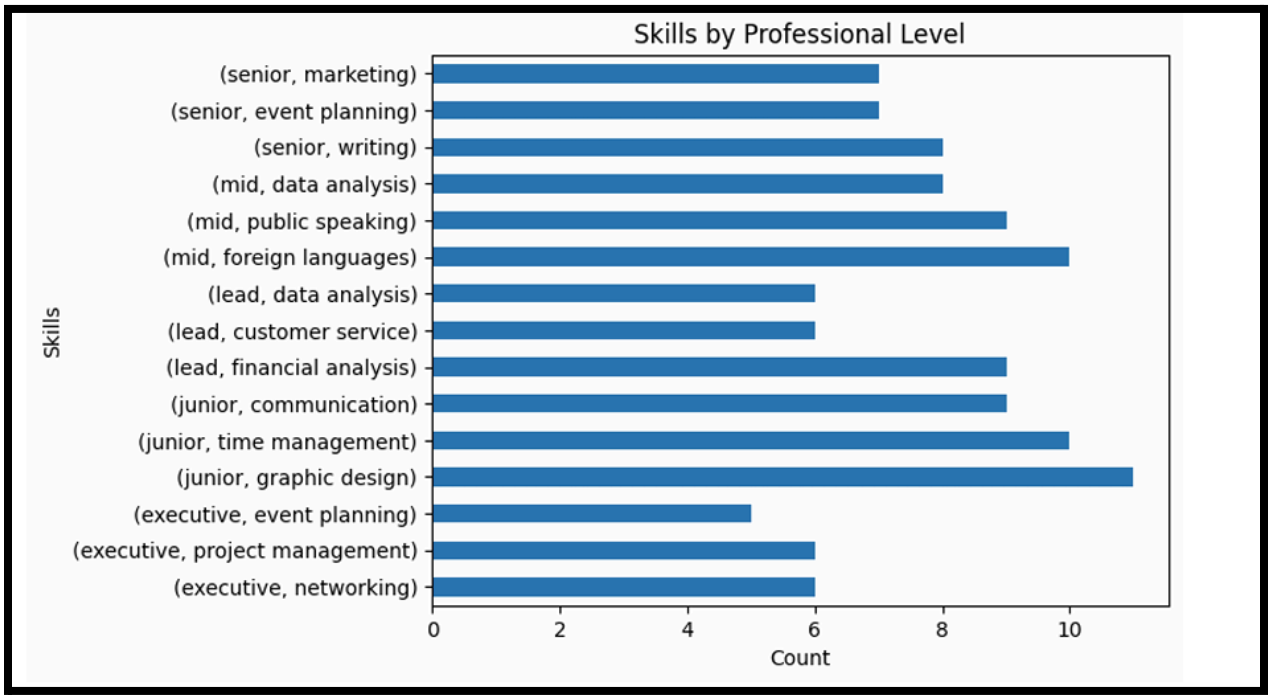
The next measures taken to uncover patterns in the data were to analyze the skills, technical proficiencies, and additional training categories as stratified by professional level.

Graph 7: Bar Chart of Skills Distribution



The above bar chart shows all the categories of skills represented in the dataset with their corresponding frequency of appearance in the resume entries. The least common skill overall is negotiation while the most common skill overall is networking. However, there do not appear to be drastic differences between skill representation.

Graph 8: Bar Chart of Top 3 Skills by Professional Level and Respective Frequency



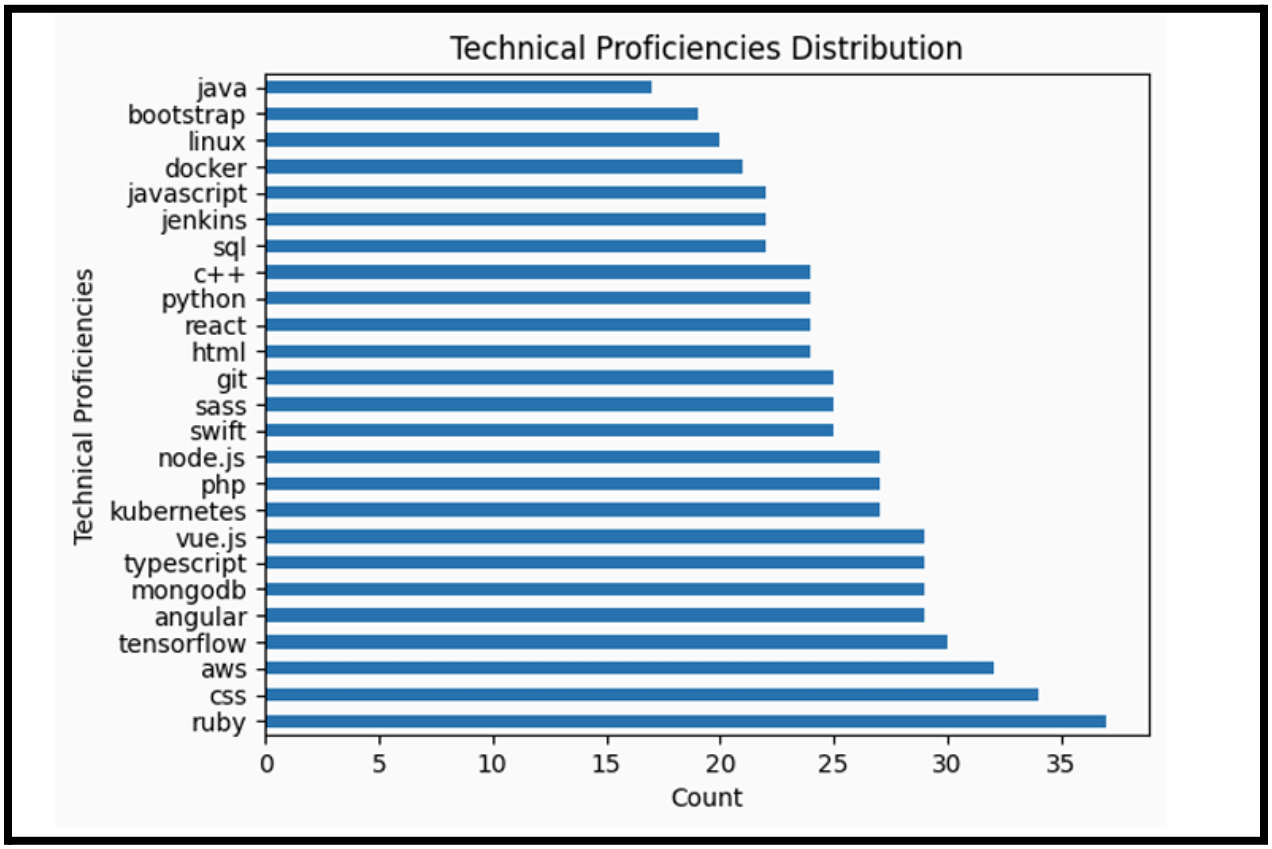
The above bar chart shows the top 3 skills by professional category and their respective frequency. A few notable trends:

1. Higher level employees (senior to executive) show higher frequency of "soft" skills related to execution and management, such as event planning, project management, and networking, showing the importance of managing complex projects and relationships for higher-level positions.
2. Lower level employees (junior to mid) show higher frequency of "hard" skills like graphic design and data analysis combined with "soft" skills related to reporting such as communication and public speaking, showing the necessity of combining these talents for entry-level roles.
3. The most frequent skills overall occur at the junior level (graphic design, 11 counts) and mid level (foreign languages, 10 counts).

The Pearson Chi-Square Test was then run to determine if there is a statistically significant difference in the distribution of skills by professional level ("Python - Pearson's Chi-Square Test"). This was performed by creating a contingency table, applying the proper functions, and analyzing the resultant chi-square statistic and p-value. The initial analysis

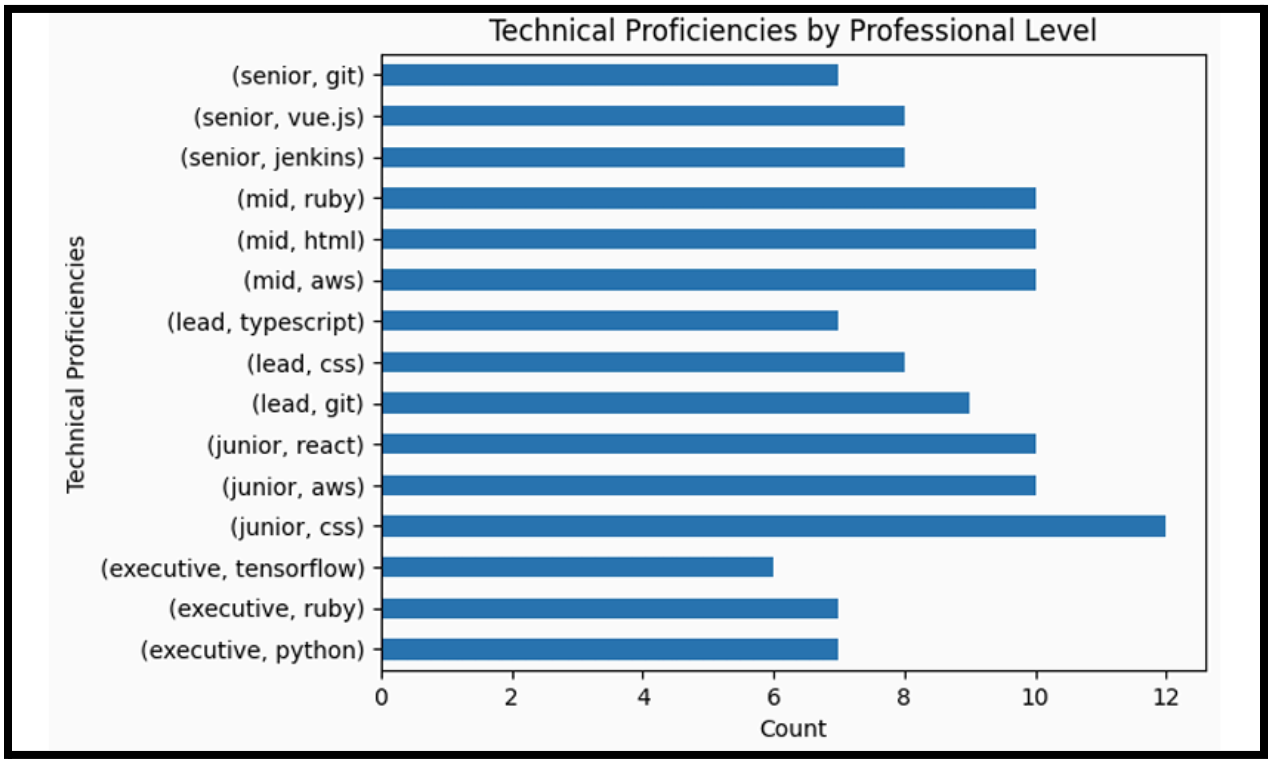
resulted in null values for the chi-square statistic and p-value. In order to correct this, which was likely due to sparse data, only skills that appeared frequently were included. However, there was no threshold number that created the necessarily-dense data, so no conclusions can be drawn regarding this relationship using this measure. Other techniques could combine the professional levels in order to create the density, but it was decided to keep this division as it was an important measure for the project.

Graph 9: Bar Chart of Technical Proficiencies Distribution



The above bar chart shows all the categories of technical proficiencies represented in the dataset with their corresponding frequency of appearance in the resume entries. The least common technical proficiency overall is Java while the most common technical proficiency overall is Ruby. However, there do not appear to be drastic differences between skill representation.

Graph 10: Bar Chart of Top 3 Technical Proficiencies by Professional Level and Respective Frequency



The above bar chart shows the top 3 technical proficiencies by professional category and their respective frequency. A few notable trends:

1. Junior to mid level employees both display high frequency of proficiency in CSS, which is used for styling HTML documents. These levels also focus on foundational technologies like AWS and HTML.
2. Executive level employees show proficiency in TensorFlow, which is more useful for advanced data and AI applications, showing the industry trends required for C-level executives.
3. The highest overall frequency is for CSS, specifically at the junior level.

The Pearson Chi-Square Test was then run to determine if there is a statistically significant difference in the distribution of technical proficiencies by professional level. This was performed by creating a contingency table, applying the proper functions, and analyzing the resultant chi-square statistic and p-value. The initial analysis resulted in null values for the chi-square statistic and p-value. In order to correct this, which was likely due to sparse data, only skills that appeared frequently were included. However, there was no threshold number that

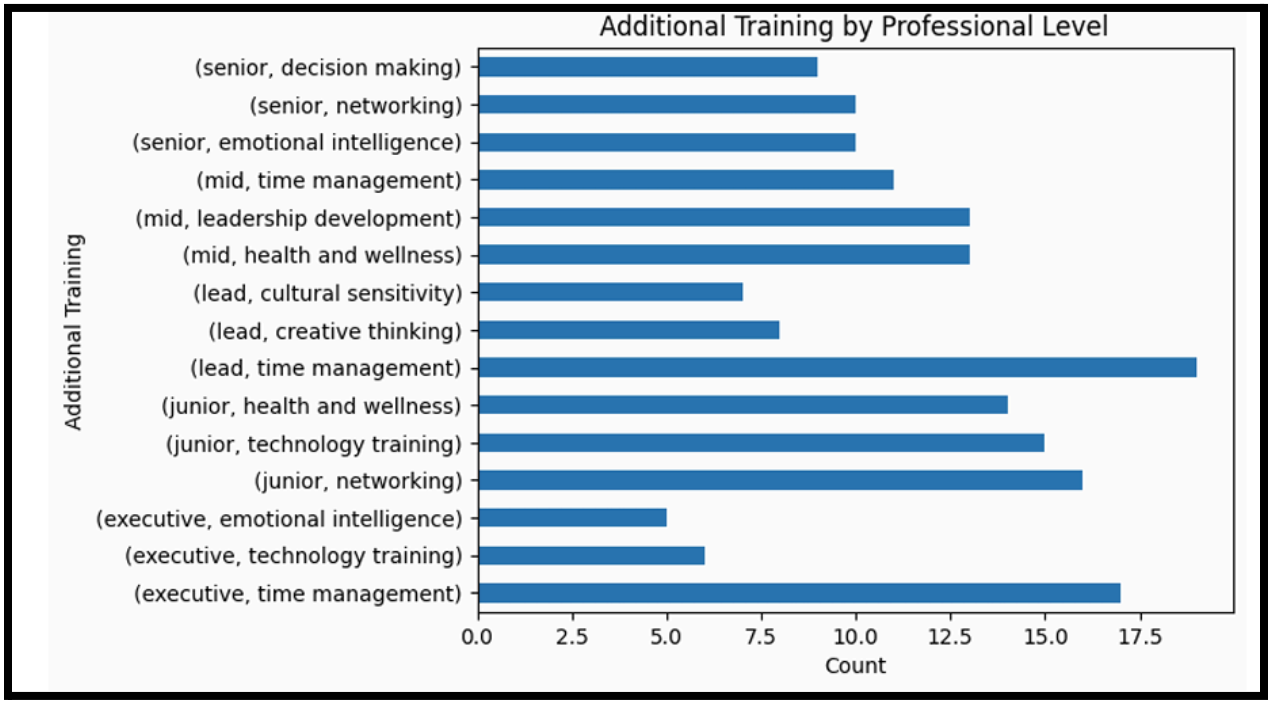
created the necessarily-dense data, so no conclusions can be drawn regarding this relationship using this measure. Other techniques could combine the professional levels in order to create the density, but it was decided to keep this division as it was an important measure for the project.

Graph 11: Bar Chart of Additional Training Distribution



The above bar chart shows all the categories of training represented in the dataset with their corresponding frequency of appearance in the resume entries. The least common training overall is sales training while the most common skill overall is time management. There seems to be very little variation within the distribution of training with the exception of the overrepresentation of time management training.

Graph 12: Bar Chart of Top 3 Additional Training by Professional Level and Respective Frequency



The above bar chart shows the top 3 additional trainings by professional category and their respective frequency. A few insights:

1. Lead and executive levels require soft skills focused on maintaining relationships and interacting with others, such as emotional intelligence and cultural sensitivity, as well as techniques for ideating and managing complexity, such as creative thinking and time management.
2. Junior and mid levels focus on individual skills and maintenance, such as technology training, health and wellness, and leadership development.
3. Time management is a crucial skill across various professional levels.

The Pearson Chi-Square Test was then run to determine if there is a statistically significant difference in the distribution of additional training by professional level. This was performed by creating a contingency table, applying the proper functions, and analyzing the resultant chi-square statistic and p-value. The analysis resulted in a chi-square statistic of appx. 98.32, with 76 degrees of freedom, and a p-value of appx. 0.044. With these results, we are able to reject the null hypothesis and interpret that additional training distributions differ by professional level.

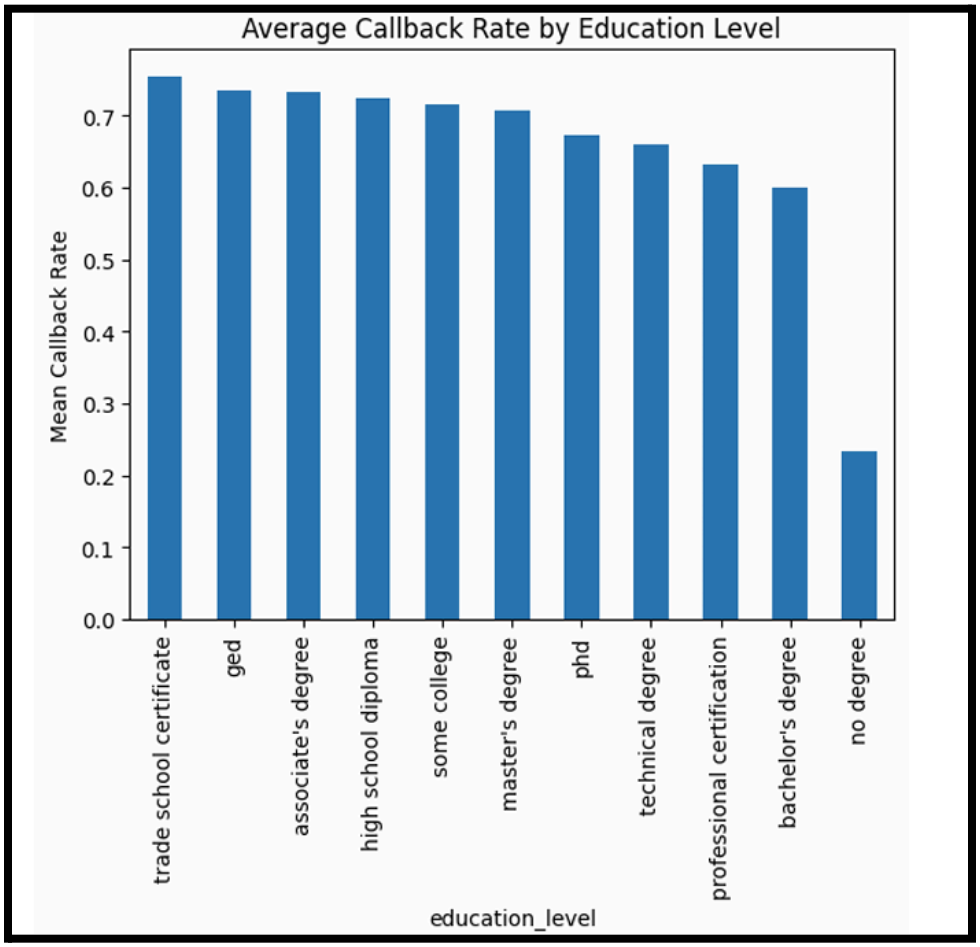
Statistical Testing: Resume Acceptance

The next set of analyses are geared towards determining what differences exist between resumes that are accepted to positions and resumes that are rejected, and what features have the most influence. First, a count of the resumes that were accepted and rejected in the analyzed dataset was performed, showing 413 accepted resumes compared to 231 rejected resumes. This shows a great overrepresentation of accepted resumes, which can influence the following analyses. Perhaps implementing a different sampling technique could help reduce this bias. These numbers also do not reveal what differences might exist.

First, the numerical attributes of years of experience, age, and GitHub contributions were analyzed for normality (using a Shapiro-Wilk test) and for statistically significant difference in distribution (using a Mann-Whitney U test, as the data for all attributes was not normally distributed, as indicated by **Graph 1** and **Graph 2**). It was determined that there is a significant difference in years of experience between accepted and rejected resumes, so experience can influence the outcome of a resume being accepted. However, different years of experience could be desired in different ranges for different positions, but it is likely to see that more experienced applicants are more likely to be accepted. It was also determined that there is no statistically significant difference in both age of applicant and number of GitHub contributions between accepted and rejected resumes, and variation seen could be due to chance. Age and GitHub contributions are then, at least by themselves, not important factors for determining acceptance. This can also be seen as an inherent flaw with the data and analytical techniques applied, as different ages or GitHub contributions could be desirable for different positions.

Second, the categorical attributes were analyzed for statistically significant differences in distribution between rejected and accepted resumes using a Chi-square test.

Graph 13: Bar Chart of Average Callback Rate by Education Level



The above chart shows the average callback rate by educational level. Applicants with a trade school certificate are the most likely to receive a callback (appx. 75.44% callback rate), while applicants with no degree are least likely to receive a callback (appx. 23.29% callback rate). There appears to be a fairly uniform distribution of callback rates with the exception of the "no degree" category. A chi-square test was then performed to determine if there was a statistically significant difference in callback rate by educational level, and the results showed a failure to reject the null hypothesis; there is no strong evidence that accepted vs. rejected resumes differ by educational level (chi-square statistic = 65.389; degrees of freedom = 10; p-value = 0).

Callback rate was calculated to determine if having references available influences callback probability. Resumes with references available had an average 65.11% callback rate, while resumes without references available had an average 63.12% callback rate. Upon initial review, these rates are almost equal. A chi-square test validated that instinct, failing to reject the null hypothesis and showing no strong evidence that resumes that are accepted/rejected differ by

having references available (chi-square statistic = 0.188; degrees of freedom = 1; p-value - 0.66427).

Callback rate was calculated to determine if professional level influences callback probability. Higher-level candidates (executive and lead) both had 100% callback rates, while lower levels (junior, mid, senior) had callback rates that were fairly uniform and appeared to be just around chance: 52.6%, 42.3%, and 49.2%, respectively. A chi-square test was then performed to determine if there was a statistically significant difference in callback rate by professional level, and the results showed a failure to reject the null hypothesis; there is no strong evidence that accepted vs. rejected resumes differ by professional level (chi-square statistic = 164.938; degrees of freedom = 4; p-value = 0).

Chi-square tests were then implemented on the remaining categorical variables to determine if there was a statistically significant difference in distribution of those categories between accepted and rejected resumes. The results are as follows:

1. No strong evidence that accepted vs. rejected resumes differ by skills (chi-square statistic = 40.934; degrees of freedom = 29; p-value = 0.06978).
2. No strong evidence that accepted vs. rejected resumes differ by languages spoken (chi-square statistic = 21.027; degrees of freedom = 24; p-value = 0.63711).
3. No strong evidence that accepted vs. rejected resumes differ by technical proficiencies (chi-square statistic = 21.933; degrees of freedom = 24; p-value = 0.58329).
4. No strong evidence that accepted vs. rejected resumes differ by additional training (chi-square statistic = 17.728; degrees of freedom = 19; p-value = 0.54067).

These results provide very little insight into the complex nature of the resumes. This can likely be attributed to the artificial nature of the data, as everything is randomly distributed, not knowing the job being applied to, and there is only one of each category represented per resume, which is not consistent with real-world resumes. Additionally, this might be a more holistic process, in which it is a combination of variables rather than one specific factor that creates the difference between acceptance and rejection.

Principal Component Analysis

Principal Component Analysis (PCA) was applied to the dataset to reduce the dimensionality introduced by extensive categorical encoding and to better understand the dominant patterns in the data ("Principal Component Analysis with Python"). The resume dataset contains both numerical variables (age, experience, GitHub contributions) and numerous categorical variables that were transformed using One-Hot Encoding, so the resulting feature space becomes high-dimensional and sparse ("Implementing PCA in Python with scikit-learn").

Before applying PCA, the dataset underwent structured preprocessing through a pipeline: Numerical features were imputed (the median) and standardized using StandardScaler, Ordinal features (education rank, professional level rank) were imputed and standardized, and Nominal features (skills, languages, technical proficiencies, additional training) were One-Hot Encoded, expanding them into dozens of binary columns. All preprocessed variables were combined through a ColumnTransformer. The PCA model was made to extract 4 principal components:

Table 4: Principal Component Analysis Results

Components
PC1: ~0.1636
PC2: ~0.1176
PC3: ~0.1120
PC4: ~0.0958

Together, these four components explained ~48.9% of the total variance in the dataset. This means that nearly half of all meaningful variation across the resume features can be summarized by just four linear combinations of variables and this compression represents substantial dimensional reduction.

Directly interpreting these components was difficult because PCA mixes numerical, ordinal, and one-hot encoded categorical variables together, each PC is a weighted linear combination of many sparse binary columns, and no single feature dominates any given

component due to standardization. Further analysis can be geared towards determining which attributes each of these principal components correlates with.

PCA was not used for prediction in the final machine learning models, but it helped with assessing data structure by displaying that almost half of the variation in the dataset can be explained by four components, meaning the feature space contains redundancy. The high number of categorical features creates an extremely sparse dataset and PCA confirms how much of that sparsity falls into shared patterns. The use of PCA also supports K-Prototype clustering, showing that the dataset groups applicants based on correlated attribute sets.

Machine Learning

The results of the ML Models ability to predict callbacks was evaluated by three main models, using Logistic Regression, the Random Forest Model, and K-Prototype Clustering.

Logistic Regression

Logistic regression is a simple, linear classification algorithm used to predict the probability of a binary outcome- in this case, whether or not a resume was accepted ("Logistic Regression using Python"). Logistic models are known for their interpretability, simplicity, efficiency, and strong baseline performance. Unfortunately, logistic models are limited by their assumption of linearity, issues with feature importance, and sensitivity to outliers.

First, the data was preprocessed using ColumnTransformer to scale numerical features with StandardScaler and convert categorical data into numerical format using One-Hot Encoding ("Using ColumnTransformer in Scikit-Learn for Data Preprocessing"). Once this pipeline was applied, a logistic regression classifier was developed, trained (80%/20% train/test split), and used to predict results. Within the code, a pre-determined random state of 42 was used as a parameter in order to ensure reproducibility of results. Additionally, the maximum number of iterations parameter was set to 1000, which is the maximum number of "passes" the algorithm will perform on the training data before deciding on its solution. This logistic regression model was appx. 66.67% accurate, meaning that it performs slightly better than chance, but still with lots of errors.

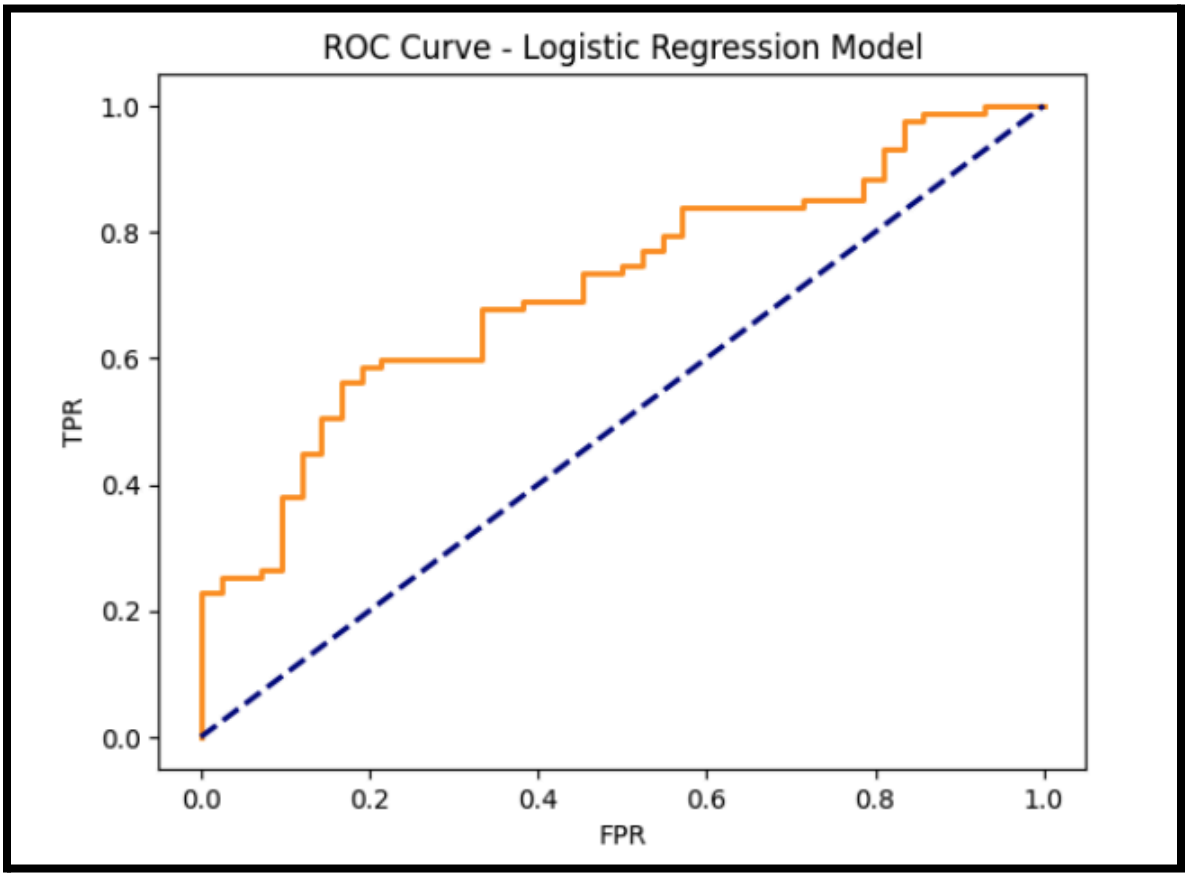
Table 5: Logistic Regression Confusion Matrix

	Predicted (0)	**Predicted (1)**
Actual (0)	20 (True Negative)	22 (False Positive)
Actual (1)	21 (False Negative)	66 (True Positive)

The Confusion Matrix shows that there is misclassification in both positive and negative classes. There is evidence that the logistic regression model has a moderate ability to identify resumes that were accepted, but it struggles with rejections. The false positives and false

negatives are almost equal values, meaning the model's accuracy (~66.7%) and ROC-AUC Score (0.70) determines its prediction is only slightly above random chance.

Graph 14: ROC Curve - Logistic Regression Model



The model achieved a ROC-AUC score of approximately 0.70, which indicates moderate predictive performance. This score is not exceptionally high, but confirms that the model is able to capture some underlying patterns in the resume features. The model is relatively effective at identifying applicants who are likely to receive callbacks when using stricter probability thresholds. Overall, the logistic regression ROC curve illustrates that the model performs better than random chance but not strongly.

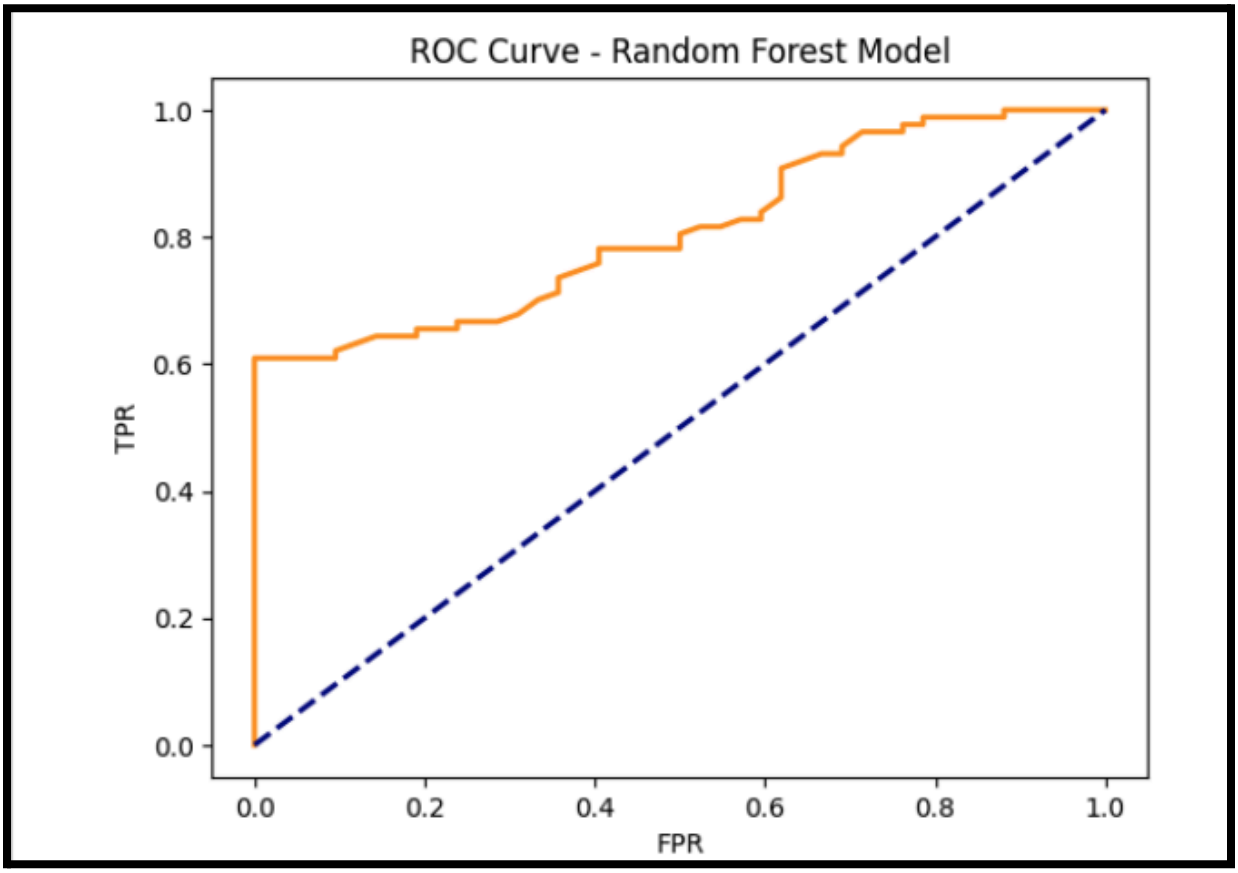
Random Forest Model

A random forest model was implemented as it is useful combining multiple decision trees for a single outcome and is useful for classification and regression. RF models output the class

that is the mode of the classes or the mean prediction of the individual trees. The random forest model was geared towards creating a more holistic analysis of the resume dataset. Random forest models are known for their accuracy, robustness, ability to handle non-linear feature relationships, built-in feature importance, and low risk of overfitting data. Unfortunately, RF models are limited by their reduced interpretability, high computational cost, and bias towards categorical data.

First, the data was preprocessed using the same ColumnTransformer as in the logistic regression model. This pipeline is used to scale numerical features with StandardScaler and convert categorical data into numerical format using One-Hot Encoding. Once this pipeline was applied, a random forest classifier was developed, trained (80%/20% train/test split), and used to predict results. Within the code, a pre-determined random state of 42 was used as a parameter in order to ensure reproducibility of results. Additionally, the number of estimators parameter was set to 200 (a fairly standard number), which is the number of individual trees that the model will construct for its bootstrap aggregating methodology. This random forest model was appx. 70.54% accurate, meaning that it is an acceptable model and able to discriminate well, but has room for improvement. Overall, the random forest model is far more successful than the logistic regression model.

Graph 15: ROC Curve - Random Forest Model



The model does a better analysis with establishing applicants who receive callbacks and applicants who don't receive one. This model has a high TPR with low FPR values, which shows that the model is effective at correctly identifying early even before false positives show. The ROC-AUC score of appx. 0.81 represents a higher level of discriminative power, meaning the Random Forest Model has significant predictive capabilities.

Feature importance scores were also extracted from the random forest model, but without the corresponding features, these numbers provide no insight. This is a topic for future analysis.

K-Prototype Clustering

With the issues presented by predictive modeling and the constraints of the artificial data, an additional form of modeling was implemented to gain insight. K-Prototype clustering was implemented in order to develop clusters of successful applicant profiles. K-Prototype techniques were used as it was most suitable for this kind of data, as it can work with various data types (Brunet).

First, a mask was applied to the training dataset to only include the resumes that were deemed accepted to the position, as the desired insight is into successful resumes. Next, a specific preprocessor was developed in order to provide StandardScaler to the numerical and ordinal columns, and use OrdinalEncoder for categorical variables. Once this preprocessor was applied to the training set, and the proper transformations were applied, a final accepted array was developed for the k-prototypes method.

The Elbow Method was used to determine the optimal number of clusters based on the cost curve, and it was determined to be 4 clusters. Before performing this technique, it was crucial to determine that the data was properly dense for the analysis; sparsity is a continuous issue with this dataset.

Finally, the K-Prototypes model was developed, fit to the data, and the cluster results were displayed in **Table 6** and **Table 7**.

Table 6: Cluster Centroids (Numerical) from K-Protoypes

Cluster	**Age**	**Experience (yrs)**	**GitHub Contributions**	**Education Rank**	**Level Rank**
0	58.0	19.6	270	2.63	4.09
1	65.8	20.7	821	5.53	3.57
2	65.7	22.2	292	8.22	3.05
3	48.5	9.5	533	5.41	1.87

Table 7: Cluster Modes (Categorical) from K-Protoypes

Cluster	Most Common Skill (Mode)	Most Common Language (Mode)	Most Common Technical Proficiency (Mode)	Most Common Additional Training (Mode)
0	problem solving	Korean	TensorFlow	Time management
1	communication	Dutch	HTML	Time management
2	writing	Thai	Jenkins	Technology training
3	graphic design	Dutch	CSS	Health and wellness

Cluster 0: Applicants in this cluster are older (avg. age ≈ 58) and have about 20 years of experience. Their education level is relatively low, but they have high professional positions (level rank ≈ 4, equivalent to Lead). Their profiles emphasize problem solving, Korean language skills, TensorFlow proficiency, and additional training in time management. This cluster represents highly experienced operational or managerial professionals with trade-skill or hands-on technical backgrounds.

Cluster 1: This is the oldest group (avg. age ≈ 66) and features extremely high GitHub contributions (mean ≈ 821), suggesting deep engagement in software engineering or open-source projects. Their education level is mid-range, but they commonly demonstrate communication skills, Dutch language knowledge, HTML proficiency, and time management training. This cluster reflects seasoned senior technologists with strong digital portfolios.

Cluster 2: Similar in age to Cluster 1 (avg. age ≈ 66) but has the highest education level (education rank ≈ 8.22, corresponding to bachelor's/master's degrees). They also have the highest average work experience (≈ 22 years). Their modal attributes include writing skills, Thai language proficiency, Jenkins technical experience, and technology training. This cluster represents academically strong senior engineers or technical specialists.

Cluster 3: This is the youngest cluster (avg. age ≈ 48.5) with the least experience (≈ 9.5 years). Their education is mid-level, and professional level is low (≈ junior/mid). They are characterized by graphic design, Dutch language skills, CSS proficiency, and health and wellness training. This cluster appears to represent mid-career technical workers or UX-oriented professionals.

Results

After cleaning the data, 644 entries were considered for further analysis, and are the data upon which these results are based. Exploratory data analysis showed abnormal distributions of age and years of experience, with both the youngest and least experienced candidates underrepresented in the dataset, while both the older and the more experienced candidates were overrepresented. Correlation tests show a weak but statistically significant positive correlation that as age increases, years of experience increase, but are unable to draw any significant correlations between age and professional level, age and educational level, and professional level and educational level.

Table 8: Skills, Technical Proficiencies, and Additional Training by Professional Level

	Junior	**Mid**	**Senior**	**Lead**	**Executive**
Skills	Graphic Design, Time Management, Communication	Foreign Languages, Public Speaking, Data Analysis	Writing, Marketing, Event Planning	Financial Analysis, Customer Service, Data Analysis	Networking, Project Management, Event Planning
Technical	CSS, AWS, React	AWS, HTML, Ruby	Jenkins, Vue.js, Git	Git, CSS, TypeScript	Python, Ruby, TensorFlow
Additional Training	Networking, Technology Training, Health and Wellness	Health and Wellness, Time Management, Emotional Intelligence	Emotional Intelligence, Networking, Decision Making	Time Management, Creative Thinking, Cultural Sensitivity	Time Management, Technology Training, Emotional Intelligence

The above table shows the most frequent skills, technical proficiencies, and additional training by professional level in order from most frequent to least frequent within each category. These results were summarized from **Graph 7, 8, 9, 10, 11, and 12**. Chi-square tests revealed no

significant differences in skill and technical proficiency distribution by professional level, but successfully rejected the null hypothesis and found statistically significant differences in additional training distribution. In **Table 8**, the skills highlighted in blue are the most common overall for the resume attribute. The least common overall skill was negotiation, technical proficiency was Java, and training was sales training. The results reveal the following trends about professional level profiles:

Table 9: Professional Level Profiles

	Lower Professional Level (junior, mid)	**Higher Professional Level (senior, lead, executive)**
Skills	Hard skills: graphic design, data analysis Soft skills: communication, public speaking *(reporting and executing specific tasks)*	Soft skills: event planning, project management, networking *(managing complex projects and relationships)*
Technical Proficiency	CSS, AWS, HTML *(foundational technologies)*	TensorFlow *(advanced data and AI applications)*
Additional Training	Technology training, health and wellness, leadership development *(individual skills and maintenance)*	Soft skills: emotional intelligence, cultural sensitivity *(interaction)* Techniques: creative thinking, time management *(ideating and managing complexity)*

Next, analysis was performed between accepted (413) and rejected (231) resumes. The analysis showed that years of experience was the only numerical variable that significantly differed between accepted and rejected resumes, while age and GitHub contributions did not show meaningful differences, and variation can be due to chance. Most categorical features such as skills, references available, languages spoken, technical proficiencies, and additional training also showed no strong independent relationship with acceptance because of the synthetic nature of the dataset.

For principal component analysis, it was found that four key features explained almost half (48.9%) of the variance in the dataset, but lacked any correspondence to the actual features. PCA confirmed that variance was spread across many features rather than dominated by a single factor.

For modeling, Logistic Regression reached about 66.7% accuracy with a ROC-AUC around 0.70, showing moderate predictive ability. The Random Forest model performed better with ~70.5% accuracy and a ROC-AUC near 0.80, meaning it captured more complex interactions between features. The K-Prototypes clustering identified four groups of accepted candidates, showing clear “candidates profiles” based on shared patterns in experience, education, and technical skills. Overall experience and combination of resume features and not single attributes were the strongest predictors of acceptance in this dataset.

Table 10: Cluster Profiles

	Cluster 0	**Cluster 1**	**Cluster 2**	**Cluster 3**
Description	*Older, experienced operational employees*	*Older, highly technical, mid-range roles*	*Older, highly educated, experienced specialists*	*Younger, novice technical employees*
Avg Age	58.0	65.8	65.7	48.5
Avg Experience (Years)	19.6	20.7	22.2	9.5
Avg GitHub Contributions	270	8212	293	533
Avg Education	2.63 (high school diploma, trade school certificate)	5.53 (professional certification, technical degree)	8.22 (Bachelor’s degree, Master’s degree)	5.41 (professional certification, technical degree)
Avg Professional Level	4.09 (lead)	3.56 (senior, lead)	3.05 (senior)	1.87 (junior, mid)
Skill-Based Characteristics	Problem solving, Korean, TensorFlow, Time management	Communication, Dutch, HTML, Time management	Writing, Thai, Jenkins, Technology training	Graphic design, Dutch, CSS, Health and wellness

Conclusion and Discussion

This project demonstrates how data cleaning, EDA, PCA, clustering, and machine learning models can be used together to understand resume patterns. While the dataset is artificial, the results highlight how hiring decisions often depend on multiple interacting features rather than one dominant variable. Experience emerged as the clearest predictor, while individual categorical variables had limited impact on their own.

The models showed that Random Forest outperformed Logistic Regression by capturing nonlinear relationships, though at the cost of interpretability. PCA and clustering helped address this by revealing the underlying structure of accepted candidates and grouping them into meaningful categories.

Because the data is synthetic, these findings should not be interpreted as real world hiring trends but they do show how combined statistical and machine learning workflow can offer insights into resume evaluation while raising important considerations about fairness and transparency.

Error Analysis

There are a variety of possible sources of error in this project. First and foremost is the artificial nature of the data, as it is randomly distributed, does not represent real-world trends, and is not structured in the same way as real-world resumes (for example, only one “skill” per resume). Even the exact measurements and definitions of the attributes were unclear and had to be interpreted. Additionally, the job acceptance labels are artificial, so no conclusive results can be applied to the real world. While various logic checks were performed in order to clean the data and make it as realistic as possible, this can only account for so much. The skew of the age and years of experience (overrepresented older/more experienced, underrepresented younger/novice) as well as the skew of resume category (413 accepted, 231 rejected) can both introduce bias into the results.

In terms of performing any form of statistical analysis, sparse data resulted in many inconclusive results, such as for statistical differences in feature distribution between accepted and rejected resumes. This can also be seen as an inherent flaw with the data and analytical techniques applied, as different ages or GitHub contributions could be desirable for different

positions. The results from these Chi-square tests provide very little insight into the complex nature of the resumes. This can likely be attributed to the artificial nature of the data, as everything is randomly distributed, not knowing the job being applied to, and there is only one of each category represented per resume, which is not consistent with real-world resumes. Additionally, this might be a more holistic process, in which it is a combination of variables rather than one specific factor that creates the difference between acceptance and rejection.

Issues with working with various types of data and technical know-how also contributed to lack of information and interpretability, for example, with not knowing the features to which the principal component analysis points.

Future Work

Future improvements could include applying this workflow to a real resume dataset with information including text fields. More advanced models such as gradient boosting or neutral networks could also improve predictive accuracy and be paired with explainability tools like SHAP. Another thing we could also incorporate fairness and bias metrics to assess whether certain groups are unintentionally disadvantaged. Finally, clustering and PCA insights could be expanded into recommendation tools, helping applicants understand which combination of skills and experience align with accepted profiles.

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